



NASA's Holoportation Achievement

In a first, NASA successfully "holoported" a Doctor from Earth to the International Space Station. <https://digit.in/may22-62>



Blockheads?

Jack Dorsey drops the title of CEO from his fintech startup Block and changed it to Block Head. <https://digit.in/may22-63>

How to install TWRP on Android

A step-by-step guide to installing TWRP custom recovery

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WHAT IS RECOVERY?

Recovery refers to the dedicated, bootable partition on which the recovery console is installed in Android. Your phone will boot to recovery mode with a combination of key presses where you can discover tools to help restore your installation as well as install official OS upgrades. Building a customised version with more options is pretty easy because Android is open and has the recovery source code available.

The stock's rebound is limited, but that's on purpose. Its primary use is to erase user's data and files, or to manually upgrade the system using signed and confirmed delta update packages. Both of these processes are normally initiated from the operating Android system, but you can do it manually and boot directly into recovery. Recovery is what boots up and erases the files and data when you tell your phone to conduct a factory reset. Similarly, when we restart to install an official OS update, we do so in recovery mode. It's quite handy, but it's also quite limiting. Different manufacturers employ various recovery methods, but they all serve the same purpose: to give the tools required to restore a functional system.

Custom Android recoveries have been updated to allow backup and restore functions, selective data deletion so you don't have to wipe

everything, and update packages. You may also mount different partitions to copy files to the SD card without having to remove it or restart Android. They are a crucial tool for anyone who wants to hack their Android phone or tablet because of the additional functionality packed in.

WHERE TO GET RECOVERY?

Typing the name of the device along with 'recovery' should be enough to land you on the page that has the right recovery for your device. The most popular recovery is TWRP or Team Win Recovery Project. They have all the devices for which official TWRP releases are put out so they do perform thorough testing of the recovery before making it available for download. Searching in Google for the recovery made for your device will either land you on some XDA forum page or on the TWRP download page. If the XDA page is for an unofficial TWRP release then do make sure to read the comments to see if it's working properly on people's devices. Otherwise, always prefer the TWRP official release.

INSTALLING TWRP ON YOUR ANDROID PHONE

Here's a step by step guide for installing TWRP on your Android phone. Remember, if you do not find an official release, don't proceed. We are not responsible for wrecking your phone.

Download Android SDK Tools

Before installing TWRP, you will need to install the Android SDK tools. On that page, if you just want the command-line tools instead of the whole Android Studio, click "Down-

load options" then select the package that corresponds to your operating system from the "Command line tools only" section of the Android developer website. Extract the command line tools to a location of your choice.

Download the TWRP Image for your device

Head over to the website to find all the official TWRP releases and download the one for your device. You can find the full list of official image files on TWRP's website. Image files or .img files are one of the types of files that can be flashed in Android.

Unlock Developer Options

Developer options on Android are fantastic, allowing you to access a variety of features and functions not available in "stock" Android, such as USB debugging, adjusting animation speeds, and a night mode that changes eligible apps to a dark theme at sunset. This method should be essentially identical from Android version 4.2 to Android 10, while there may be minor differences based on the brand and UI of your phone.

Go to "Settings -> About Phone" to enable developer settings. (Settings -> System -> About phone, for example.)

Repeat tapping "Build number" until you see a notice that says "You're now a developer."

To check the developer options are enabled, you'll need to enter your PIN code. Return to "Settings -> System -> Advanced," where the new "Developer options" will be shown.

However, be cautious because certain categories, such as "Running services," contain items that are critical to the operation of your phone.

Enable USB Debugging

Within developer options, you will find an option that reads Enable USB Debugging or plainly USB Debug-